

University of Southern Denmark

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Written Exam

AI503 – Calculus

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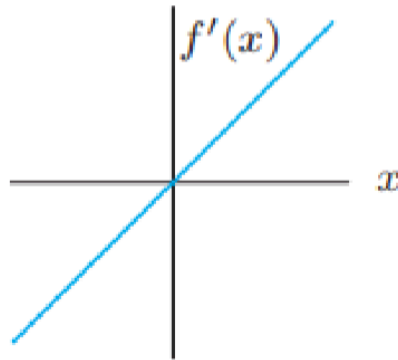
Duration:	2 hours
Allowed aids:	All written materials
Date:	March 2nd 2026



DEPARTMENT OF MATHEMATICS
AND COMPUTER SCIENCE

Problem 1

Use the following figure about $f'(x)$ to answer the questions:



Solution

(a) Over what intervals is f increasing? Decreasing?

- increasing in $x > 0$
- decreasing in $x < 0$

(b) Does f have local maximum or local minimum? If so, which and where?

f is a polynomial likely something like x^2 with a local minimum at $x = 0$

Problem 2

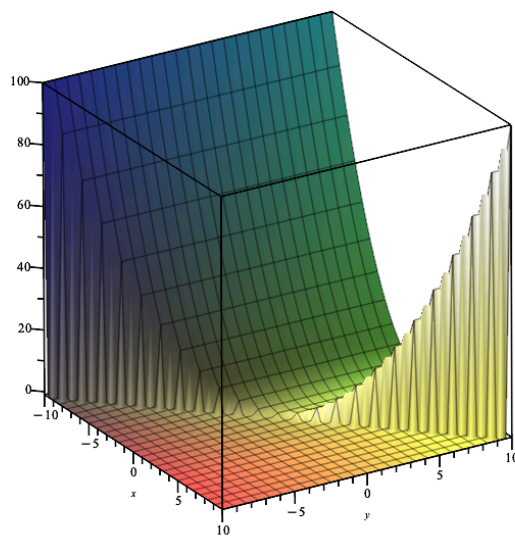
Solution

(a) Determine whether the following limit exists. It does

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2} = -\frac{1}{2} \cdot \frac{1}{2}$$

(b) Is the function continuous along $y = x$

No, there are massive jumps



Problem 3

A plane has equation $z = 5x - 2y + 7$.

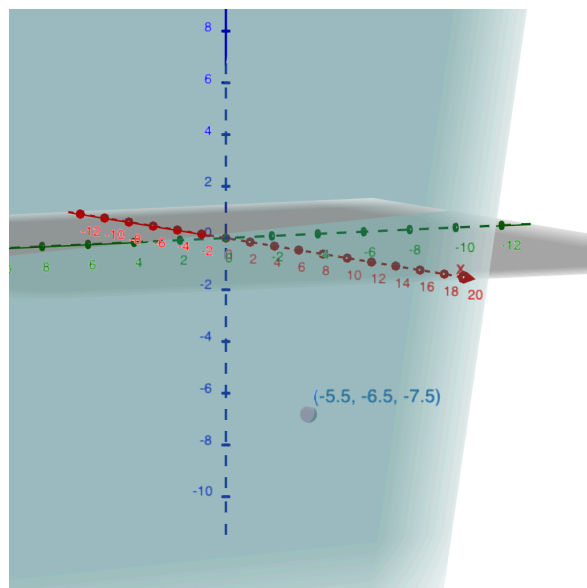
Solution

(a)

NOT done 😞

(b) Find a value of a so that the point $(a + 1, a, a - 1)$ lies on the plane.

$$a - 1 = 5(a + 1) - 2a + 7 \Rightarrow a = -6.5$$



Problem 4

Solution

Tangent of point $(1,0,9)$ is

$$0 = 3x + 6 + y - z$$

Problem 5

Solution

(a) What is the rate of change of $f(x, y) = 3xy + y^2$ at the point $(2, 3)$ in the direction $(3, -1)$?

Directional derivative is $7y + 3x$ at $(2,3)$ that is 5

(b) What is the direction of maximum rate of change of f at $(2, 3)$?

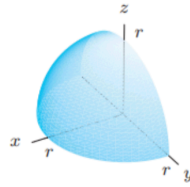
That is $\nabla f([2, 3]) = 9, 12$

(c) What is the maximum rate of change?

$$[3y, 2x + 2y]$$

Problem 6

We want to express the volume of the quarter sphere below as an integral of the spherical coordinate $\int_W f \, dV$ where W is the quarter sphere and $dV = \rho^2 \sin(\phi) \, d\rho \, d\phi \, d\theta$.



Solution

(a) What must the function f be? Function must be $f = 1$

(b) Write the limits of the integration.

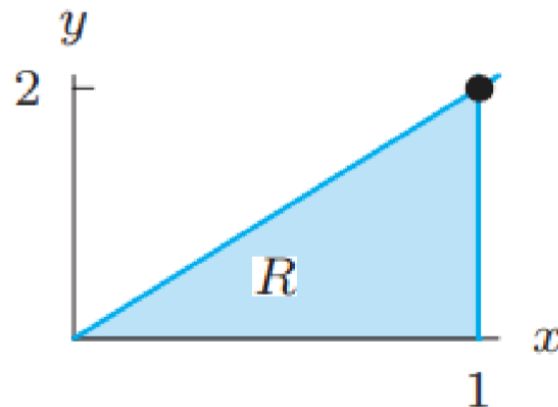
$$\int_{\theta=0}^{\frac{\pi}{2}} \int_{\phi=-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_{\rho=-\frac{\pi}{2}}^{\frac{\pi}{2}}$$

(c) Evaluate the integral.

$$\int_{\theta=0}^{\frac{\pi}{2}} \int_{\phi=-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_{\rho=-\frac{\pi}{2}}^{\frac{\pi}{2}} 1 \, d\rho \, d\phi \, d\theta = \frac{\pi^3}{2}$$

Problem 7

Integrate the function $f(x, y) = x + y$ over the shaded region below.



Solution

$$\int_0^2 \int_0^1 x + y \, dx \, dy = \frac{17}{3}$$

Problem 8

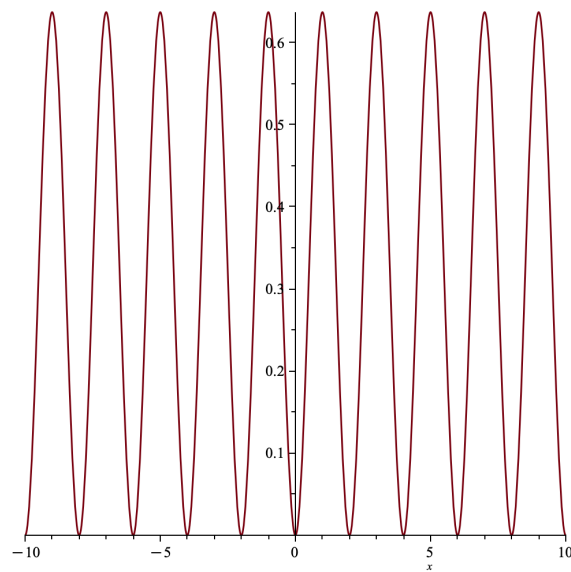
Let $F(x) = \int_0^x \sin(\pi t) \, dt$

Solution

(a) Evaluate $F(1)$

$$F(1) = \int_0^1 \sin(\pi * t) \, dt = \frac{2}{\pi}$$

(b) For what values of x is $F(x)$ positive? negative?



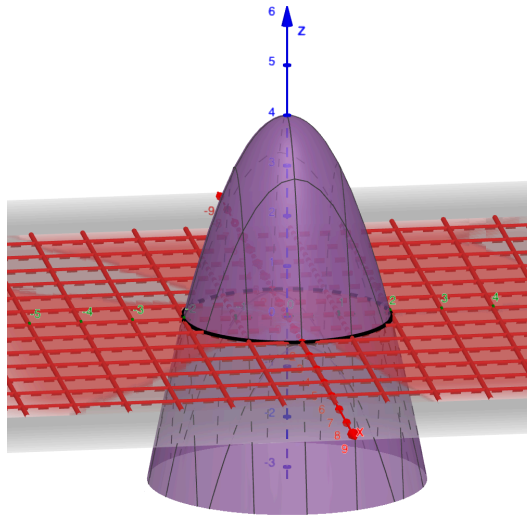
Aways $0 \leq F(x)$

Problem 9

Find the volume of the solid that lies under the paraboloid $z = 4 - x^2 - y^2$ and above the xy -plane.

Solution

Plot it



Then

$$\int_0^5 \int_{-3}^{-3} \int_{-3}^{-3} 4 - x^2 - y^2 \, dx \, dy \, dz = 360$$

Problem 10

Decide whether the following statements are true or false.

Solution

- (a) TRUE
- (b) FALSE
- (c) TRUE
- (d) TRUE
- (e) FALSE